EAST WEST UNIVERSITY

PROJECT REPORT

Designing a Full-Fledged Network for an Organization with Multiple Subnets

Submitted to:

Instructor: Dr. Anisur Rahman

Designation: Associate Professor

Department: Computer Science & Engineering

Submitted by:

Name: Nurtaz Reza

ID Number: 2023-3-60-282

Department: Computer Science and Engineering

Course Details:

Course Title: Computer Networks

Course Code: CSE405

Section: 2

Semester: Summer 2026

Date of Submission: 30th August, 2026

Table of Contents

1. Introduction 4

2. Purpose of the Network 4

3. Tools and Components 5

3.1 Software Used 5

3.2 Logical Components Used 5

4. Network Overview 5

5. Physical Network Diagram 6

6. IP Addressing and Subnet Design 7

6.1 Campus LAN Networks 7

6.2 Router-to-Router Serial Networks 8

6.3 Wireless Networks 8

7. Router Configuration 9

7.1 IP Configuration Code 9

7.2 Routing Configuration Code 12

8. DHCP Server Configuration 14

9. DNS Server Configuration 16

10. Web Server Configuration 18

11. Wireless Network Configuration 24

12. Connectivity Testing 26

12.1 Same-Network Connectivity 26

12.2 Different-Network Connectivity 26

12.3 DNS and Web Connectivity 26

13. Number of Hosts 27

14. Number of Networks 27

15. Design Issues and Challenges 28

16. Limitations 28

17. Future Expansion 29

18. Conclusion 29

1. Introduction:

The project presents the design and implementation of a multi-campus network for the University of Scholars using Cisco Packet Tracer. It connects seven campuses through routers, with each campus supporting wired and wireless devices. The network integrates subnetting, DHCP, DNS, HTTP, wireless networking, static routing, and OSPF to ensure communication between different network segments. It uses addresses from Class A, Class B, and Class C private address ranges to organize different network segments. A centralized server segment provides essential DHCP, DNS, and Web services across the organization.

2. Purpose of the Network:

The primary purpose of this project is to design a functional and scalable organizational network in which users from different campuses can communicate and access common network services.

The objectives are:

- To design seven campus LANs interconnected through routers.
- To establish communication between hosts in the same and different campuses.
- To provide both wired and wireless connectivity.
- To implement OSPF dynamic routing on Routers 2, 3, 4, 5, and 7.
- To implement static routing on Routers 1 and 6.
- To use serial links for router-to-router communication.
- To use addresses from Class A, Class B, and Class C private address ranges.
- To provide automatic IP configuration through DHCP.
- To deploy a DNS server for domain-name resolution.
- To deploy a Web server containing the Scholarsbd university website.
- To make the university website accessible using: http://abc.scholarsbd.edu
- To provide sufficient addressing space for network operation and future expansion.

3. Tools and Components:

3.1 Software Used:

Cisco Packet Tracer (v9.0.0)

3.2 Logical Components Used:

- 7 PT Routers
- Cisco switches
- Wireless routers/Home routers
- Desktop PCs
- Laptops
- Smartphones
- Tablet
- DHCP Server
- DNS Server
- Web Server
- Serial DCE/DTE connections
- Copper straight-through cables
- Wireless connections

4. Network Overview:

The University of Scholars network consists of seven interconnected campus routers, with Campus 7 serving as the central communication point. Each campus has a dedicated LAN with wired and wireless connectivity through switches and wireless routers. Campus 5 hosts the centralized DHCP, DNS, and Web servers.

5. Physical Network Diagram:

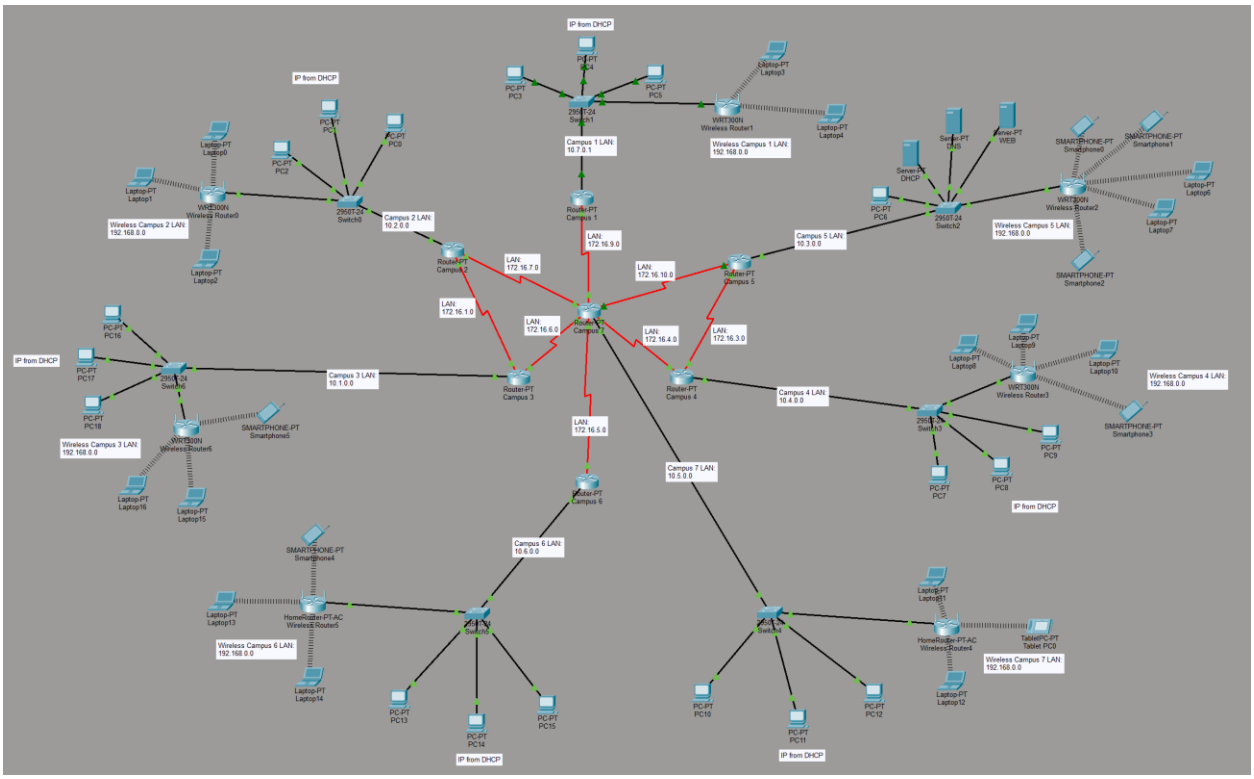


Figure: Complete Physical Network Model in Cisco Packet Tracer

The physical topology contains seven routers, multiple switches, wired and wireless hosts, and centralized network services. Serial connections are used between campus routers. Individual LAN and serial network labels are displayed on the topology to make the addressing structure easier to identify.

6. IP Addressing and Subnet Design:

6.1 Campus LAN Networks:

The campus LANs use /24 networks with subnet mask:

```
255.255.255.0
```

The configured LAN networks are:

```
Campus 1
Network: 10.7.0.0/24
Gateway: 10.7.0.1

Campus 2
Network: 10.2.0.0/24
Gateway: 10.2.0.1

Campus 3
Network: 10.1.0.0/24
Gateway: 10.1.0.1

Campus 4
Network: 10.4.0.0/24
Gateway: 10.4.0.1

Campus 5
Network: 10.3.0.0/24
Gateway: 10.3.0.1

Campus 6
Network: 10.6.0.0/24
Gateway: 10.6.0.1

Campus 7
Network: 10.5.0.0/24
Gateway: 10.5.0.1
```

Using /24 LANs leaves a substantial number of host addresses available beyond the devices currently deployed, providing room for future expansion.

6.2 Router-to-Router Serial Networks:

The serial links use addresses from the Class B range.

```
Campus 1 ↔ Campus 7
172.16.9.0/24

Campus 2 ↔ Campus 7
172.16.7.0/24

Campus 3 ↔ Campus 7
172.16.6.0/24

Campus 4 ↔ Campus 7
172.16.4.0/24

Campus 5 ↔ Campus 7
172.16.10.0/24

Campus 6 ↔ Campus 7
172.16.5.0/24

Campus 2 ↔ Campus 3
172.16.1.0/24

Campus 4 ↔ Campus 5
172.16.3.0/24
```

Serial DCE interfaces were configured with a clock rate of: `64000`

6.3 Wireless Networks:

Wireless routers provide private wireless LAN connectivity to laptops, smartphones, and the tablet. The local wireless segments use Class C private addressing, such as:

```
192.168.0.0/24
```

Wireless clients obtain appropriate IP configuration from their corresponding wireless network configuration and use the configured DNS service for access to the university website.

7. Router Configuration:

7.1 IP Configuration Code:

Campus 1 / Router 1

```
interface fa0/0
ip address 10.7.0.1 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 172.16.9.1 255.255.255.0
no shut
do wr
exit
```

Campus 2 / Router 2

```
interface fa0/0
ip address 10.2.0.1 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 172.16.7.1 255.255.255.0
no shut
do wr
exit

interface se3/0
ip address 172.16.1.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
```

Campus 3 / Router 3

```
interface fa0/0
ip address 10.1.0.1 255.255.255.0
no shut
do wr
exit
```

```
interface se2/0
ip address 172.16.6.1 255.255.255.0
no shut
do wr
exit

interface se3/0
ip address 172.16.1.2 255.255.255.0
no shut
do wr
exit
```

Campus 4 / Router 4

```
interface fa0/0
ip address 10.4.0.1 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 172.16.4.1 255.255.255.0
no shut
do wr
exit

interface se3/0
ip address 172.16.3.1 255.255.255.0
no shut
do wr
exit
```

Campus 5 / Router 5

```
interface fa0/0
ip address 10.3.0.1 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 172.16.3.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se3/0
ip address 172.16.10.1 255.255.255.0
```

```
no shut
do wr
exit
```

Campus 6 / Router 6

```
interface fa0/0
ip address 10.6.0.1 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 172.16.5.1 255.255.255.0
no shut
do wr
exit
```

Campus 7 / Router 7

```
interface fa0/0
ip address 10.5.0.1 255.255.255.0
no shut
do wr
exit

interface se2/0
ip address 172.16.9.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se3/0
ip address 172.16.10.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se4/0
ip address 172.16.6.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se5/0
ip address 172.16.4.2 255.255.255.0
```

```
clock rate 64000
no shut
do wr
exit

interface se6/0
ip address 172.16.5.2 255.255.255.0
clock rate 64000
no shut
do wr
exit

interface se7/0
ip address 172.16.7.2 255.255.255.0
clock rate 64000
no shut
do wr
exit
```

7.2 Routing Configuration Code:

Static Routing Configuration:

Campus 1 / Router 1

```
ip route 0.0.0.0 0.0.0.0 172.16.9.2
```

Campus 6 / Router 6

```
ip route 0.0.0.0 0.0.0.0 172.16.5.2
```

Campus 7 / Router 7 — Static Routes toward Router 1 & Router 6 LANs

```
ip route 10.7.0.0 255.255.255.0 172.16.9.1
ip route 10.6.0.0 255.255.255.0 172.16.5.1
```

OSPF Routing Configuration:

Campus 2 / Router 2

```
router ospf 1
network 10.2.0.0 0.0.0.255 area 0
```

```
network 172.16.7.0 0.0.0.255 area 0
network 172.16.1.0 0.0.0.255 area 0

exit
```

Campus 3 / Router 3

```
router ospf 1
network 10.1.0.0 0.0.0.255 area 0
network 172.16.6.0 0.0.0.255 area 0
network 172.16.1.0 0.0.0.255 area 0

exit
```

Campus 4 / Router 4

```
router ospf 1
network 10.4.0.0 0.0.0.255 area 0
network 172.16.4.0 0.0.0.255 area 0
network 172.16.3.0 0.0.0.255 area 0

exit
```

Campus 5 / Router 5

```
router ospf 1
network 10.3.0.0 0.0.0.255 area 0
network 172.16.10.0 0.0.0.255 area 0
network 172.16.3.0 0.0.0.255 area 0

exit
```

Campus 7 / Router 7

```
router ospf 1
network 10.5.0.0 0.0.0.255 area 0
network 172.16.9.0 0.0.0.255 area 0
network 172.16.7.0 0.0.0.255 area 0
network 172.16.6.0 0.0.0.255 area 0
network 172.16.4.0 0.0.0.255 area 0
network 172.16.10.0 0.0.0.255 area 0
network 172.16.5.0 0.0.0.255 area 0
redistribute static subnets

exit
```

8. DHCP Server Configuration:

A centralized DHCP server is deployed within the server LAN. The server is configured with a static address:

```
DHCP Server IP: 10.3.0.2
Subnet Mask:    255.255.255.0
Default Gateway: 10.3.0.1
DNS Server:     10.3.0.3
```

Seven DHCP pools were created to provide addresses to the seven campus LANs:

```
Campus 1
Gateway: 10.7.0.1
Start IP: 10.7.0.10
DNS: 10.3.0.3

Campus 2
Gateway: 10.2.0.1
Start IP: 10.2.0.10
DNS: 10.3.0.3

Campus 3
Gateway: 10.1.0.1
Start IP: 10.1.0.10
DNS: 10.3.0.3

Campus 4
Gateway: 10.4.0.1
Start IP: 10.4.0.10
DNS: 10.3.0.3

Campus 5
Gateway: 10.3.0.1
Start IP: 10.3.0.10
DNS: 10.3.0.3

Campus 6
Gateway: 10.6.0.1
Start IP: 10.6.0.10
DNS: 10.3.0.3

Campus 7
Gateway: 10.5.0.1
Start IP: 10.5.0.10
DNS: 10.3.0.3
```

All campus pools use:

```
Subnet Mask: 255.255.255.0
Maximum Users: 100
```

DHCP relay is configured on the LAN interfaces of remote campus routers using:

```
ip helper-address 10.3.0.2
```

For example:

```
interface fastethernet0/0
ip helper-address 10.3.0.2
```

Campus 5 does not require DHCP relay for its local wired hosts because the DHCP server resides in the same LAN.

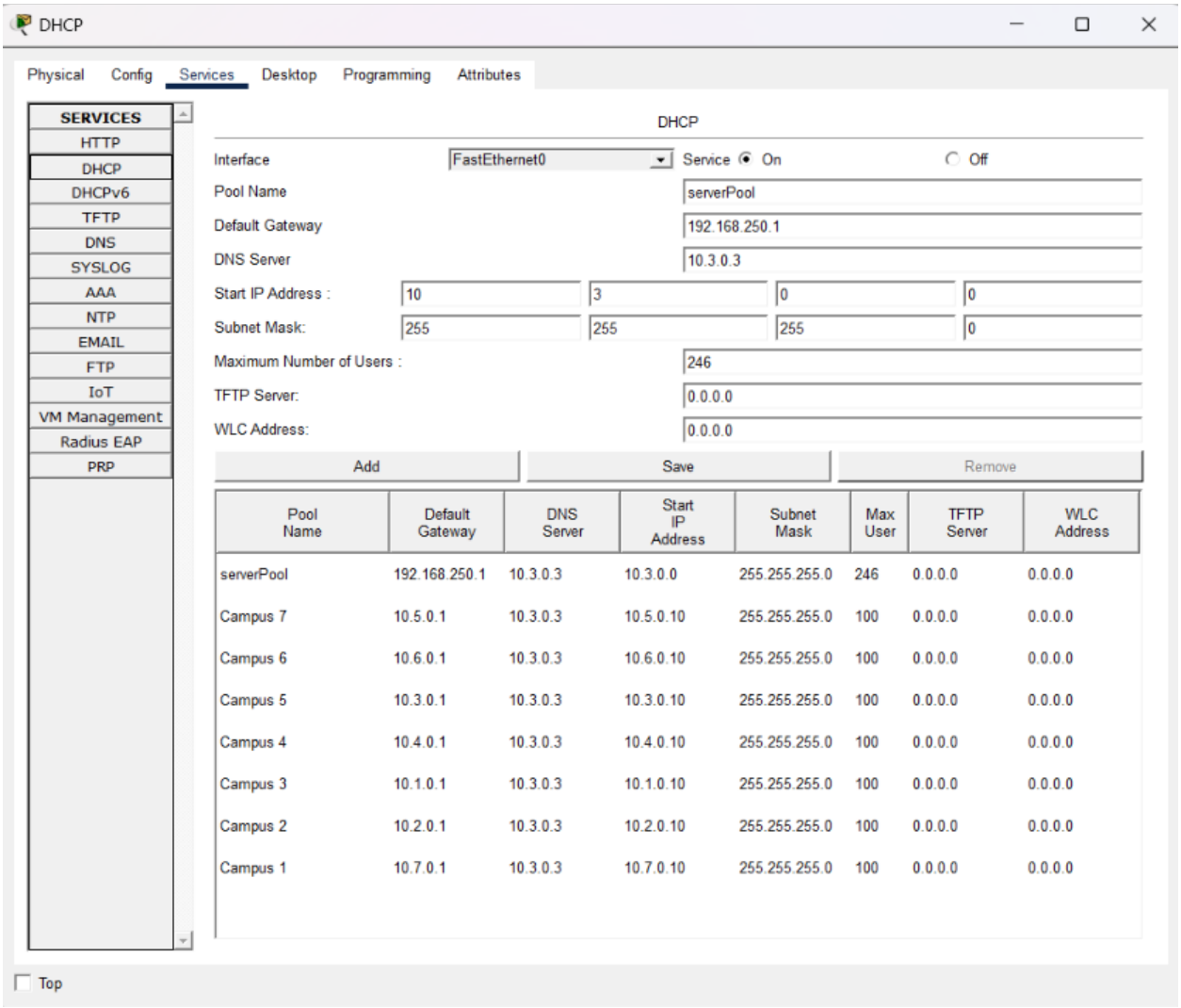


Figure: DHCP Server Pool Configuration

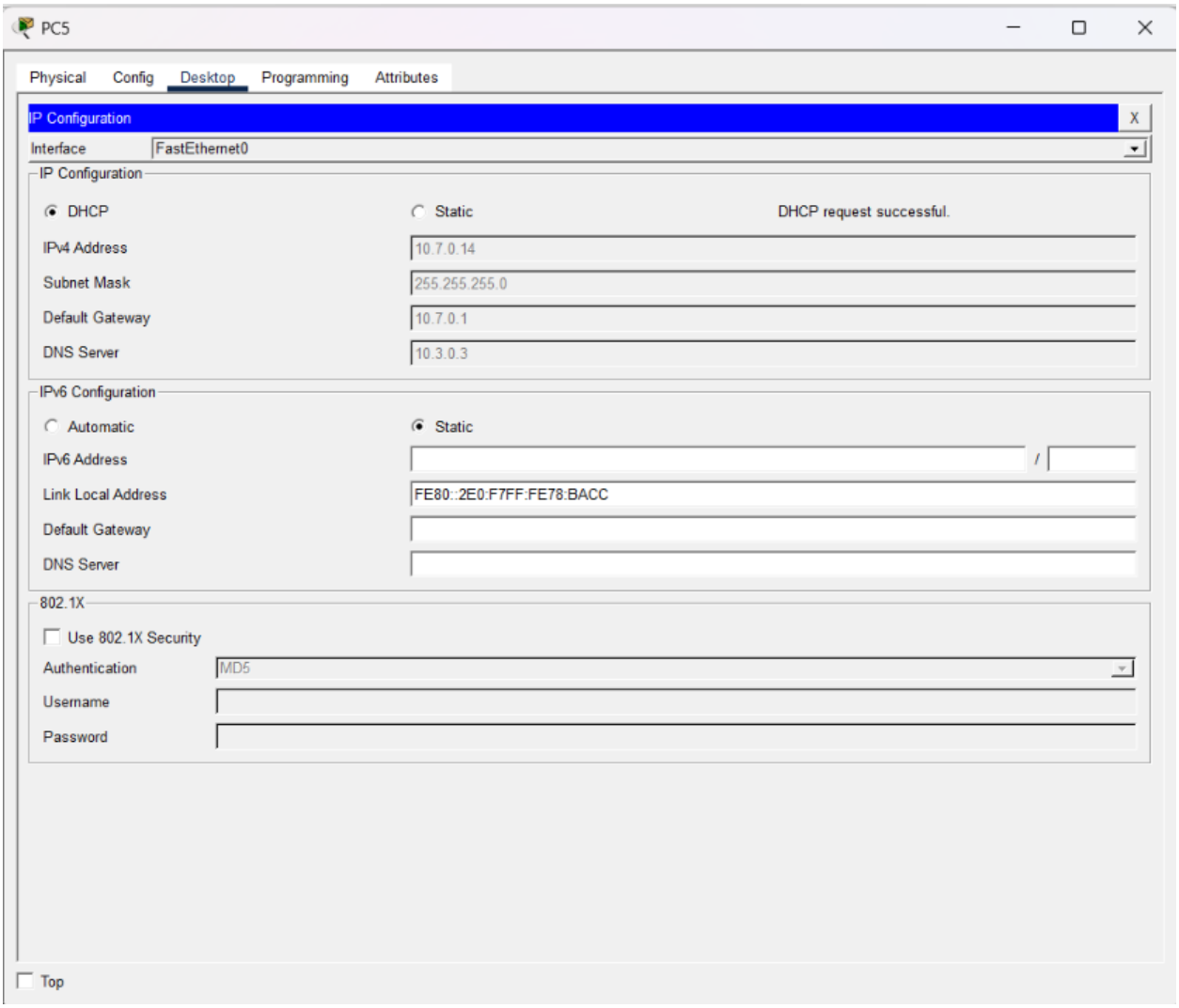


Figure: Successful Automatic IP Assignment through DHCP

9. DNS Server Configuration:

A single DNS server is configured with the following static network settings:

```
DNS Server IP:  10.3.0.3
Subnet Mask:    255.255.255.0
Default Gateway: 10.3.0.1
```

The DNS service contains an A record that maps the required university domain to the Web server:

```
Name:    abc.scholarsbd.edu
Type:    A Record
Address: 10.3.0.4
```

Therefore, users can access the university Web server through:

```
http://abc.scholarsbd.edu
```

instead of remembering its numerical IP address.

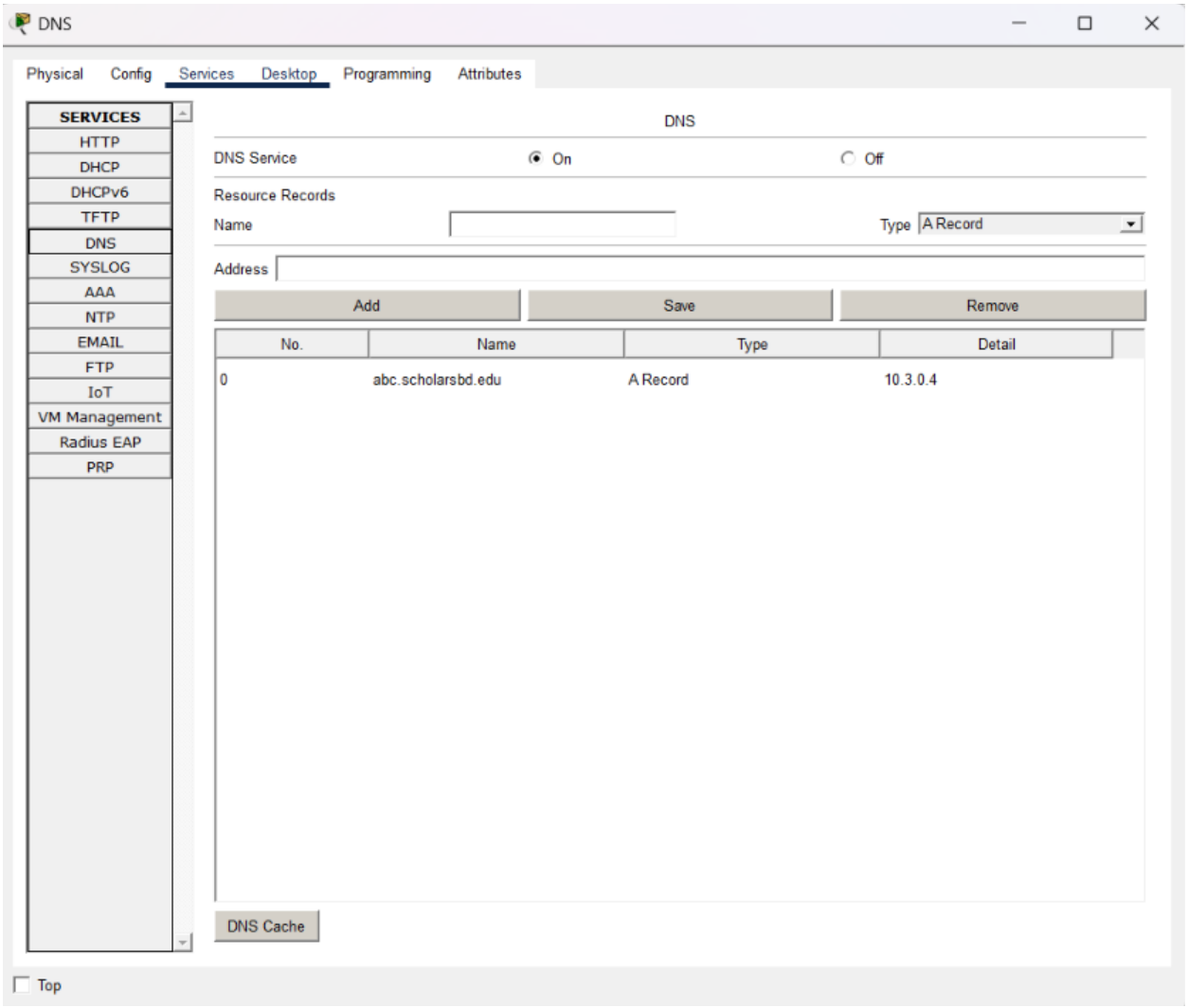


Figure: DNS A Record for abc.scholarsbd.edu

10. Web Server Configuration:

The Web server uses a static IP address:

```
Web Server IP:   10.3.0.4
Subnet Mask:     255.255.255.0
Default Gateway: 10.3.0.1
DNS Server:      10.3.0.3
```

The HTTP service is enabled on the server. A customized Scholarsbd website was created using HTML and CSS.

The website includes:

- A professional Scholarsbd home page.
- Book-themed university logo.
- Departments page.
- Admission page.
- Undergraduate admission page.
- Graduate admission page.
- Diploma admission page.
- About Us page.
- Contact section.
- Contact form with submission confirmation page.
- Consistent navigation between the pages.

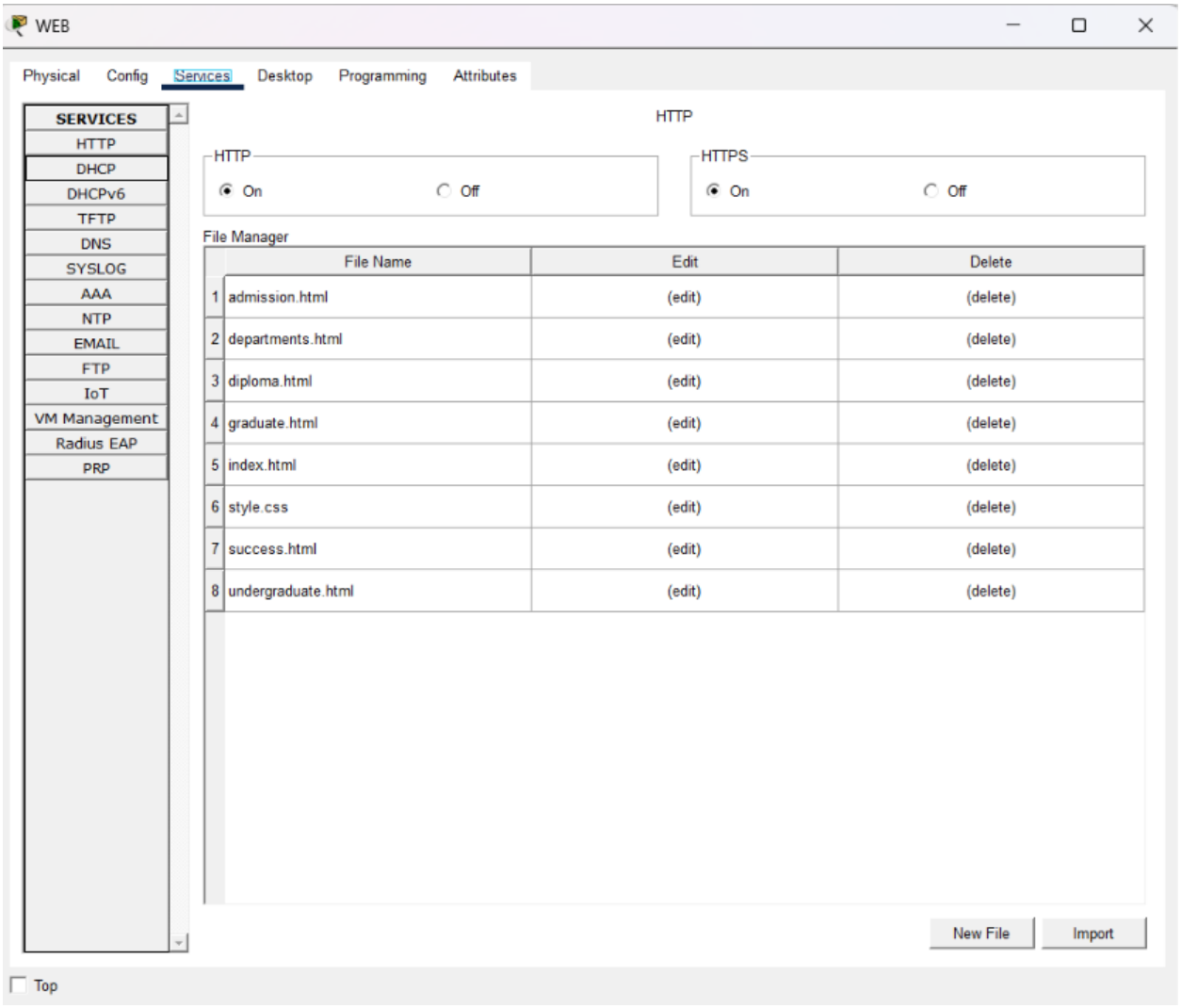


Figure: Web Server HTTP Files

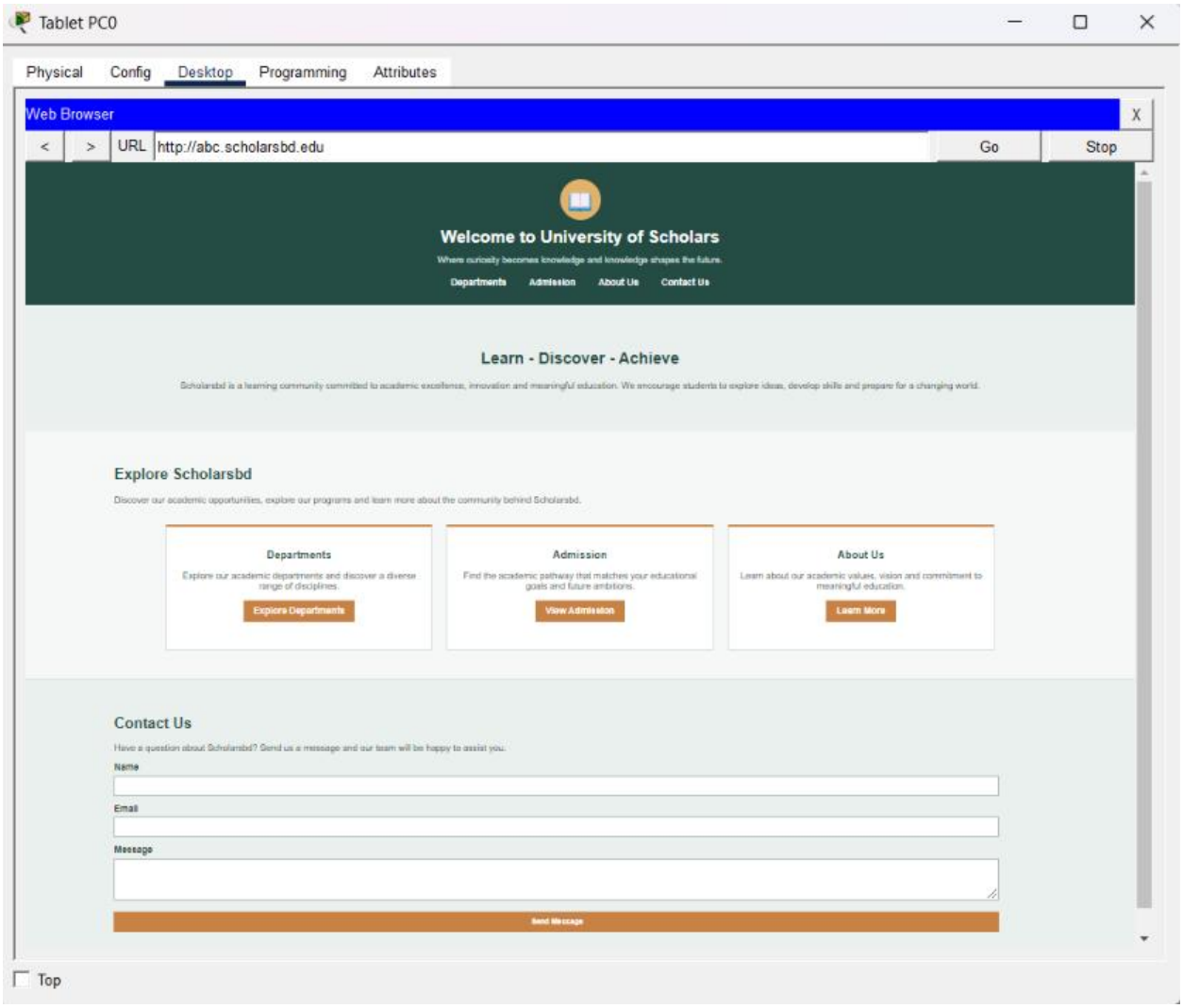


Figure: Scholarsbd University Home Page

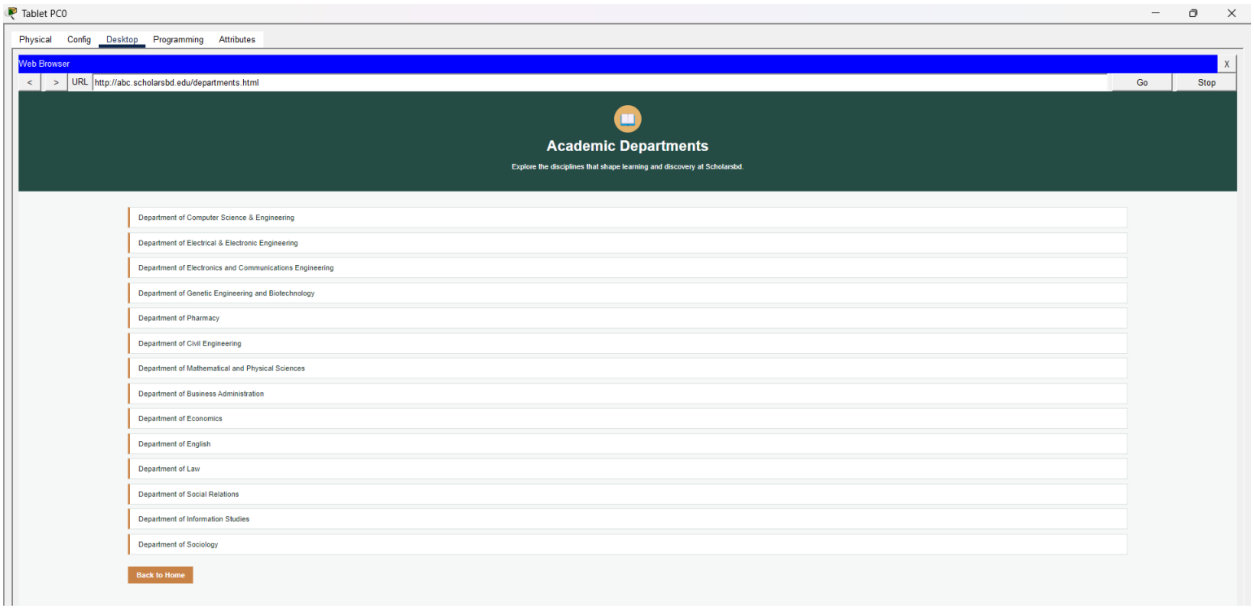


Figure: Scholarsbd Departments Page

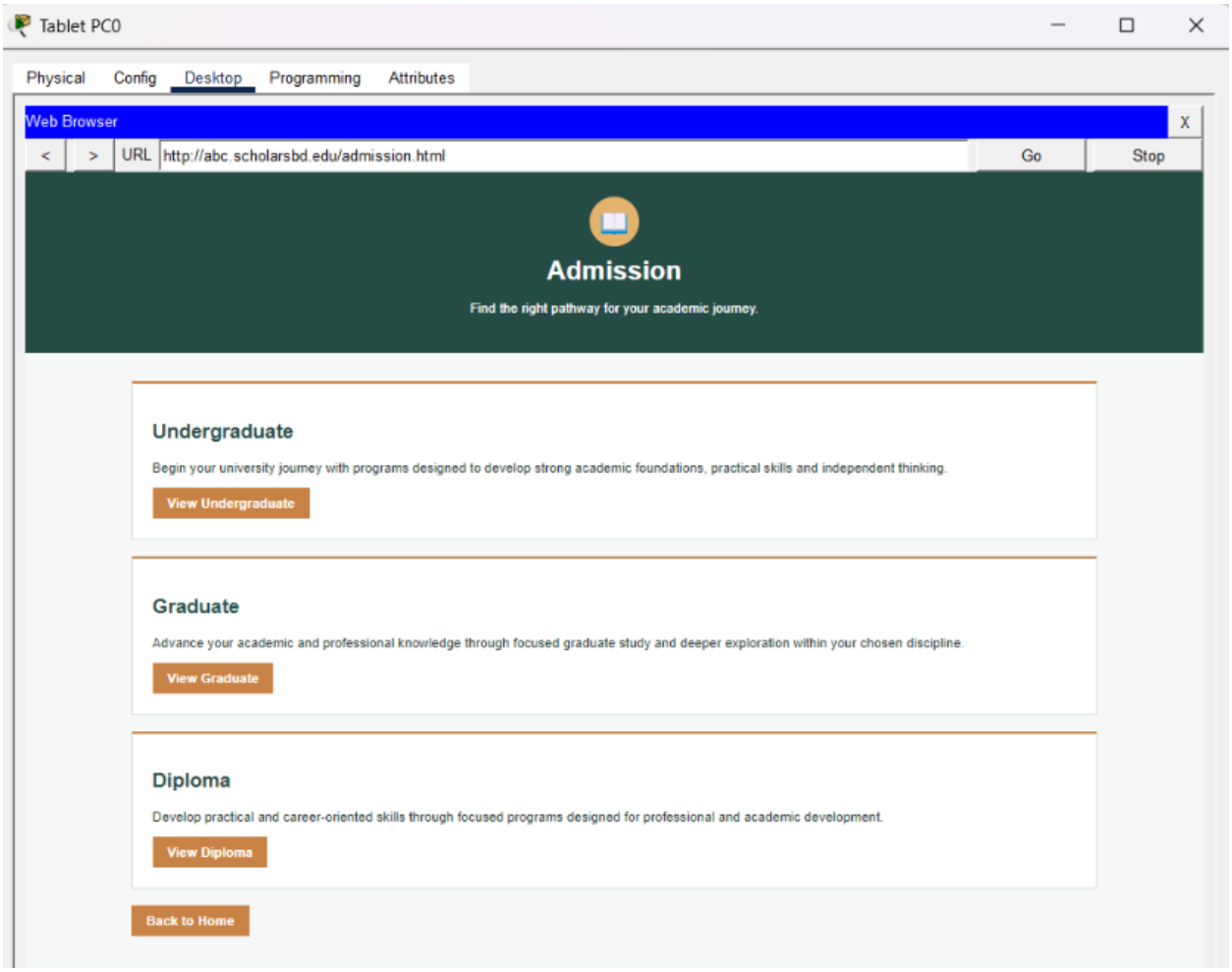


Figure: Scholarsbd Admission Page

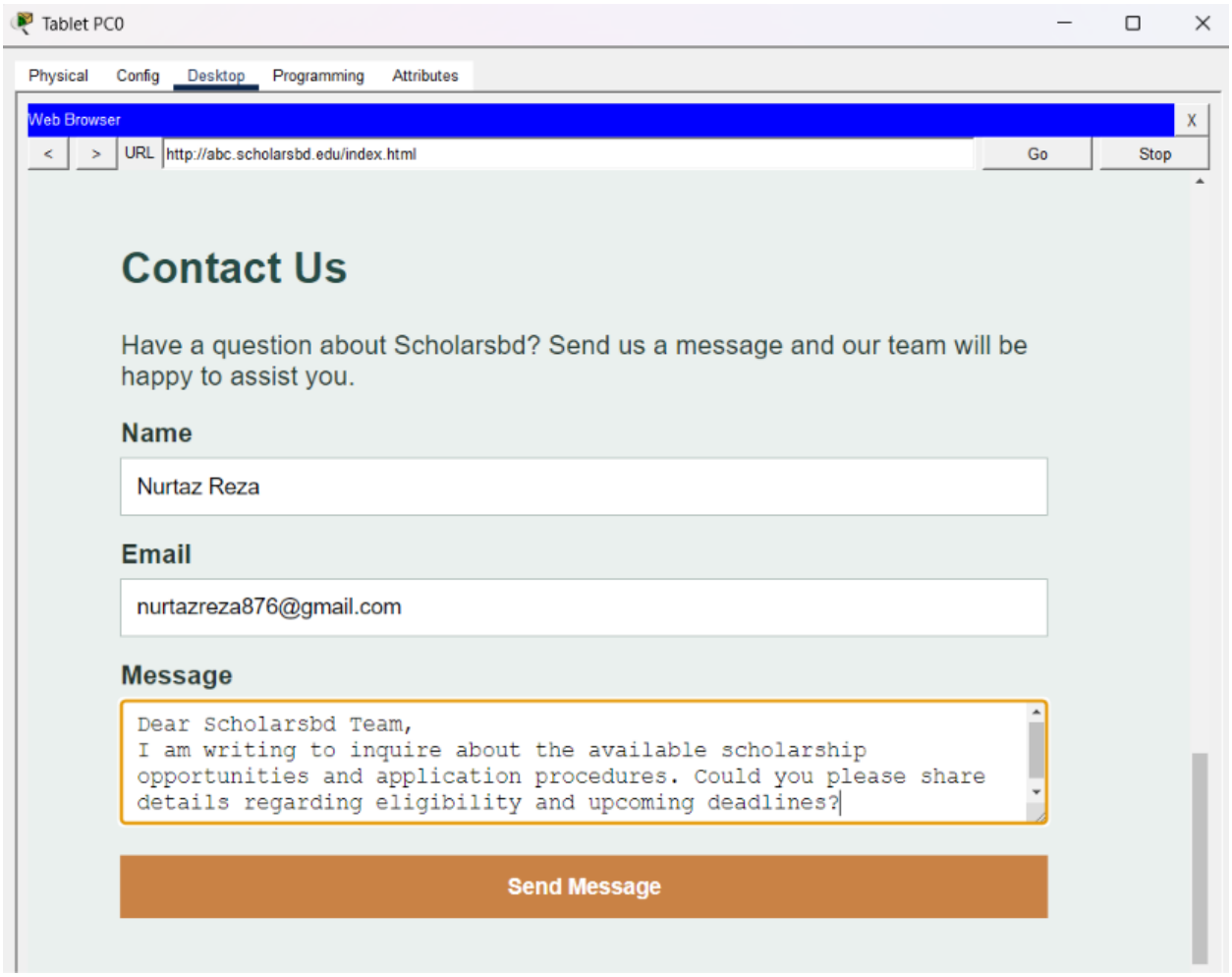


Figure: Scholarsbd Website Contact Form

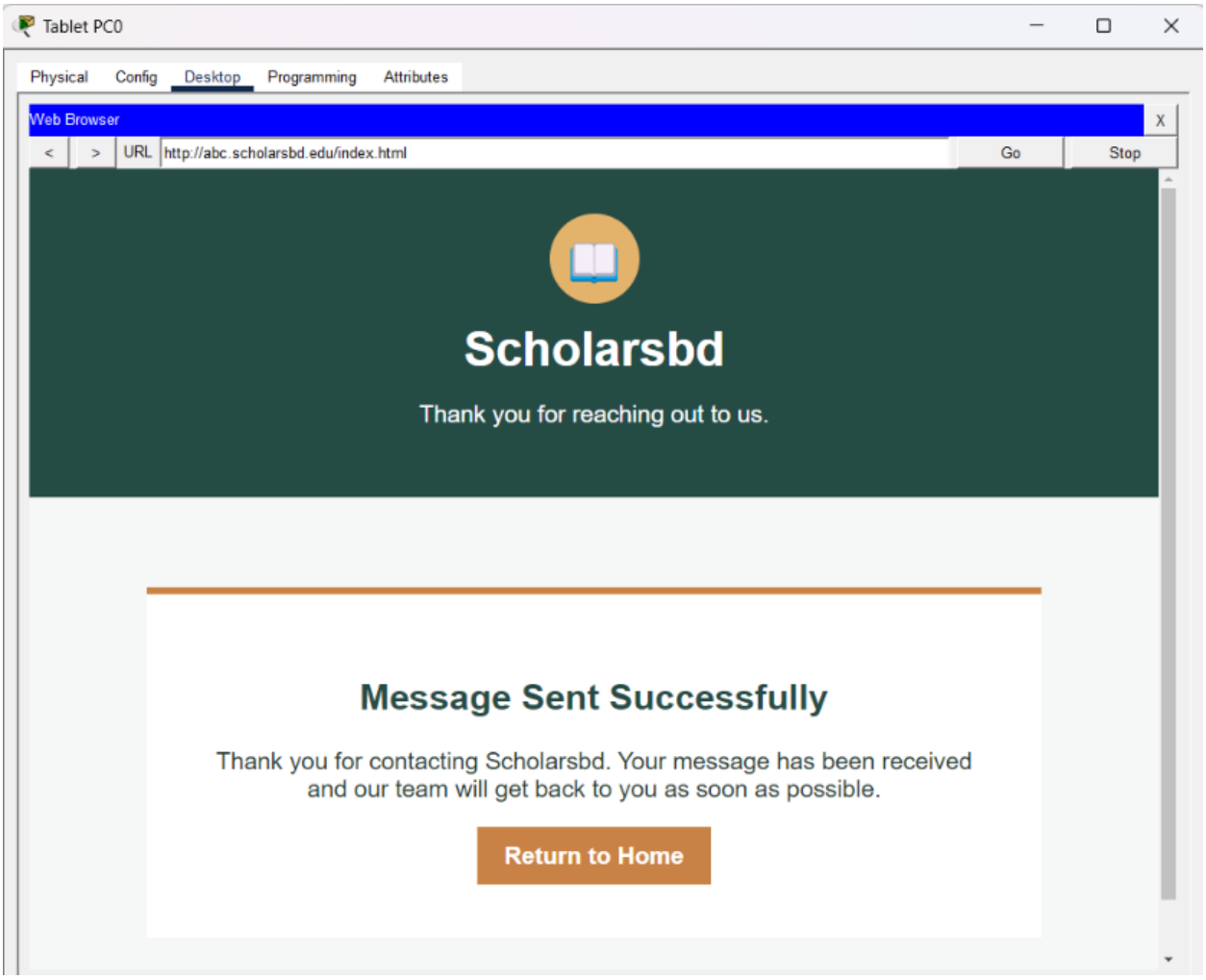


Figure: Confirmation of Successful Contact Message Submission

11. Wireless Network Configuration:

Wireless connectivity is provided on the campus networks through wireless routers. Laptops, smartphones, and a tablet are incorporated as wireless end devices.

Unique SSIDs are used where necessary to ensure that wireless clients associate with the intended campus wireless router.

The wireless devices receive appropriate network configuration and use:

```
DNS Server: 10.3.0.3
```

to resolve the Scholarsbd website.

Wireless connectivity was verified by loading:

```
http://abc.scholarsbd.edu
```

from wireless clients.

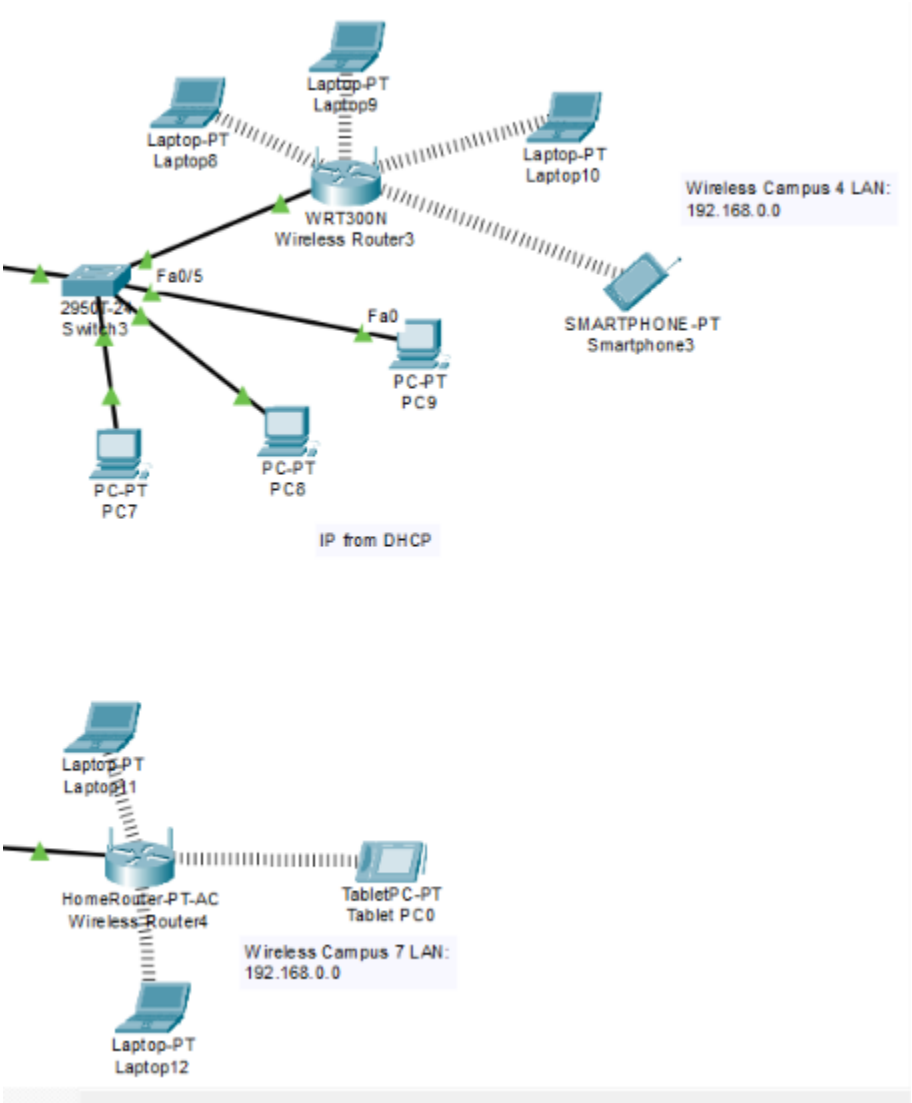


Figure: Wireless Device Connectivity

12. Connectivity Testing:

Network communication was verified using ICMP echo requests and Packet Tracer Simple PDU testing.

12.1 Same-Network Connectivity:

Two hosts located within the same campus LAN were selected as source and destination. The ICMP packet was delivered successfully, demonstrating proper switch-level and local IP connectivity.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC0	ICMP		0.000	N	0	(edit)	(delete)
	Successful	Laptop0	Laptop1	ICMP		0.000	N	1	(edit)	(delete)

Figure: Successful ICMP Communication within the Same Network

12.2 Different-Network Connectivity:

A host from one campus was used to communicate with a host located in another campus. Successful ICMP delivery confirmed that static routing, OSPF, and inter-router serial communication were operating correctly.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC5	PC6	ICMP		0.000	N	0	(edit)	(delete)
	Successful	Smart...	Smartpho...	ICMP		0.000	N	1	(edit)	(delete)

Figure: Successful ICMP Communication between Different Campus Networks

12.3 DNS and Web Connectivity:

Clients were also tested using:

```
ping 10.3.0.3
ping 10.3.0.4
ping abc.scholarsbd.edu
```

Successful responses verified access to the DNS server, Web server, and domain-name resolution.

The website was subsequently accessed using:

```
http://abc.scholarsbd.edu
```

from different parts of the network.

13. Number of Hosts:

Total number of Hosts: 42

The implemented topology contains multiple categories of end devices, including:

- Desktop PCs
- Laptops
- Smartphones
- Tablet
- DHCP server
- DNS server
- Web server

14. Number of Networks:

Total number of Networks: 22

At the routed campus level, the topology contains:

```
7 campus LAN networks
8 router-to-router serial networks
```

Therefore:

```
15 primary routed networks
```

In addition, wireless routers maintain Class C private wireless LAN segments for their associated wireless hosts.

This addressing scheme provides network separation while demonstrating the use of Class A, Class B, and Class C private address ranges.

15. Design Issues and Challenges:

Several practical challenges were encountered during implementation. These included configuring serial interfaces and DCE clock rates, integrating OSPF with static routing, and using DHCP relay through the **ip helper-address** command. Wireless devices also required correct SSIDs, connectivity, DHCP, and DNS settings. During testing, delays in ARP, DNS, and ICMP resolution were observed, but communication worked normally after the network information was established.

16. Limitations:

The implemented network has several limitations:

- Packet Tracer provides a simulated environment and therefore cannot reproduce every behavior of a production enterprise network.
- The use of /24 point-to-point serial networks reserves significantly more addresses than required; smaller point-to-point subnets would be more address-efficient in a production design.
- Wireless routers introduce additional local network configuration and management overhead.
- Additional campuses would require further addressing, routing, and physical connectivity.
- The current topology does not implement advanced enterprise security technologies such as firewall policies, ACL-based segmentation, AAA authentication, or intrusion prevention.
- Network redundancy and high-availability mechanisms could be improved in a production implementation.
- The Web service is limited by the HTTP capabilities provided by Cisco Packet Tracer.

17. Future Expansion:

The network design reserves sufficient addressing space within each /24 campus subnet to accommodate additional end devices.

Future improvements may include:

- VLAN-based departmental segmentation.
- Dedicated management networks.
- Access Control Lists (ACLs).
- Network security and authentication mechanisms.
- Redundant routers and links.
- More efficient point-to-point subnetting.
- IPv6 addressing.
- Additional servers and services.
- Enhanced wireless security.
- Expanded Web-based university services.

18. Conclusion:

The project successfully demonstrates the design and implementation of a functional multi-campus university network using Cisco Packet Tracer. Seven campus LANs were interconnected using a hybrid routing architecture with OSPF and static routing, while wired and wireless devices provided network access. Centralized DHCP, DNS, and Web services were also integrated. Connectivity and service tests confirmed that the network operates successfully across different campuses.